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VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Professional Ethics and Human Values: Ethics through Ethical Theories

Class Instructors:

Dr Chakravarthula Kiran, Prof G Srinivasa Gupta, Ms A Sandhya Rani

Unit 2

Syllabus

- Moral Development: Kohlberg's Theory vs. Gilligan's Theory; Understanding Biases
- Ethical Theories: Classification of Ethical Theories—Prescriptive, Descriptive, and Applied Ethics by Character and by Conduct; Deontological Ethics (Immanuel Kant, John Rawls), Virtue Ethics (Aristotelianism), Consequentialism (Act & Rule Utilitarianism); Divine Command Ethics and Ethics through Spirituality; Ethical Pluralism; Overview of Rights Theories
- Ethical Guidance: Ethical Frameworks and Values for organisations; Practicable guidance from the Bhagavad-Gita and Eastern Philosophies; Ethical Inquiries— Normative, Conceptual, and Factual; Ethical Decision-making Tools; Ethics and Law— Ethics as a Basis for Law, Awareness of Rights of Convicts and Criminals

Kohlberg's & Gilligan's Theories

Theories of Moral Development leading to Ethics

Kohlberg's Theory

- Three levels of moral development – Based on reasoning and motivation of individuals in response to moral questions – through social interaction
  - Pre-conventional
    - Right Conduct: Direct benefit to oneself
    - Motivation: Obedience, Desire to avoid punishment, satisfy one's needs, influence of power
    - Seen in young childrenConventional
    - Right Conduct: Acceptable by law and authority; Loyalty is most important
    - Motivation: To please, satisfy, get approved by others, and meet expectations of Society
    - Seen in adults
  - Post-conventional
    - Right Conduct: Principled conscience, no self-interest, welfare of others, moral integrity, Golden Rule, …
    - Motivation: Original thinking

Kohlberg's Theory: Six Stages

Pre-conventional

Punishment-obedience orientation

- Fear of authority and avoidance of punishment are reasons for behaving morally

Personal reward orientation

- Satisfying personal needs determines moral choice

Conventional

Good boy-nice girl orientation

- Maintaining the affection and approval of friends and relatives motivates good behaviour

Law and order/ authority orientation

- A duty to uphold rules and laws for their own sake justifies moral conformity

Post-conventional

Social contract orientation

- We obey rules because they are necessary for social order, but rules can be changed if there were better alternatives

Morality of individual principles and conscience

- Behaviour which conforms to internal principles (justice and equality) and may sometimes violate society's rules

Designed from content taken from slides titled "Moral Reasoning and Moral Development" made by Ms Jennifer Beller at Maryville University

Gilligan's Theory (Compare to Kohlberg's)

- Three levels of moral development (similar to but modified from Kohlberg's)
  - Pre-conventional: Same as Kohlberg's
  - Conventional: Not hurting others, willing to sacrifice one's (self-) interests in helping others
  - Post-conventional: Balancing care for others and self-interests, through context-oriented reasoning and not through hierarchy of rules
- "Ethics of Care" vs. "Ethics of Rules and Rights"

Gilligan's Theory: 3 Levels & Transitions



Designed from content taken from slides titled "Moral Reasoning and Moral Development" made by Ms Jennifer Beller at Maryville University

Kohlberg Vs. Gilligan

Kohlberg's Theory

- Based on study on men
- Men give importance to moral rule
- Ethics of rules and rights

Gilligan's Theory

- Based on study on women
- Personal relationship is prioritized, with inclusion of studies on women
- Ethics of care, context-oriented (Importance to circumstances leading to critical situations rather than rules)

Kohlberg Vs. Gilligan (Contd.)

Kohlberg's Theory	Gilligan's Theory
Justice	Reason
Factual	Emotional
Right or Wrong	Impact on relationships
Logic only	Logic & Compassion
Logical & Rule-based	Caring & Concern
Less of caring	More of caring
Matter of fact (Practical)	Abstract

Kohlberg Vs. Gilligan (Contd.)

Kohlberg's Theory

- Present focus
- Strict rules
- Independence
- Rigid
- Taking a commanding role
- Transactional approach

Gilligan's Theory

- Present and Future focus
- Making exceptions
- Dependence
- Human-oriented
- Shying away from decision-making
- Transformational approach

Kohlberg Vs. Gilligan: Heinz's Dilemma

- What happened?
  - Heinz, being poor and a debtor, could not buy medicine at a high cost. Pharmacist did not agree for discount of payment in instalments. Heinz stole the medicine.
- Kohlberg's Theory applied:
  - Men at Conventional level: Theft was wrong – violation of property right!
  - Men at post-conventional level: Theft was right – human life is more important.
- Gilligan's Theory applied:
  - Women: Theft was wrong – Alternate ways may have been possible
    - Context-oriented decision, not based on rules ranked in the order of priority

Watch the video below

https://www.youtube.com/watch?v=ALE5H9Bymso

Understanding Biases

Understanding Biases

- Bias: disproportionate weight in favour of or against an idea or thing, usually in a way that is inaccurate, closed-minded, prejudicial, or unfair
  - Can be natural or acquired

- https://www.youtube.com/watch?v=wEwGBIr_RIw3
- https://www.youtube.com/shorts/WeyfwNqy-bI - AI and Biases

https://en.wikipedia.org/wiki/Bias

Ethical Theories: Motivation & Overview

How Ethical Theories are created and classified and how they may help

Principles of Ethical Action

Self-interests	Never take any action that is not in the long-term self-interests of yourself and the organization to which you belong.
Personal Virtues	Never take any action that is not honest, open, and truthful, and which you would not be proud to see reported widely in national newspapers and on network television.
Religious Injunctions	Never take any action that is not kind, and that does not build a sense of community, a sense of all of us working together for a commonly accepted goal.
Govt. Requirements	Never take any action that violates the law, for the law represents the minimal moral standards of our society.
Utilitarian Benefits	Never take any action that does not result in greater good than harm for the society of which you are a part.
Rules	Never take any action that you would not be willing to see others, faced with the same or a closely similar situation, also be encouraged to take.
Individual Rights	Never take any action that abridges the agreed-upon rights of others.
Economic Efficiency	Always act to maximize profits subject to legal and market constraints and with full recognition of external costs, for maximum profits under those conditions are the sign of the most efficient production.
Distributive Justice	Never take any action in which the least among us are harmed in some way.
Contributing Liberty	Never take any action that will interfere with the rights of others for self-development and self-fulfilment.

Hosmer, L. T. "Moral Leadership in Business", Irwin (1994)

Applied Ethics

- Utilitarian Ethics bring out Applied Ethics – How do we take moral knowledge and put it into practice?
  - Decision ethics: Ethical theories and ethical decision processes
  - Professional ethics: Ethics to improve professionalism
  - Clinical ethics: Ethics to improve basic health needs
  - Business ethics: Individual-based morals to improve ethics in a business environment
  - Organisational ethics: Ethics among organizations
  - Social ethics: Ethics among nations and as one global unit
  - Technology ethics, Bioethics, Medical ethics, Computer ethics, Environmental ethics, …

https://en.wikipedia.org/wiki/Applied_ethics

Action-Oriented Ethics

- Ethics that help one decide what to do than simply what/how to be
- Emphasis is on the action and not on character
  - Different from “Virtue Ethics” – emphasis is on being in a certain way or possessing certain quality of character

Motivation & Criteria

Motivation

- To understand moral dilemmas
- To provide practical moral guidance
- To justify professional obligations and decisions
- To relate professional and personal morality

Criteria

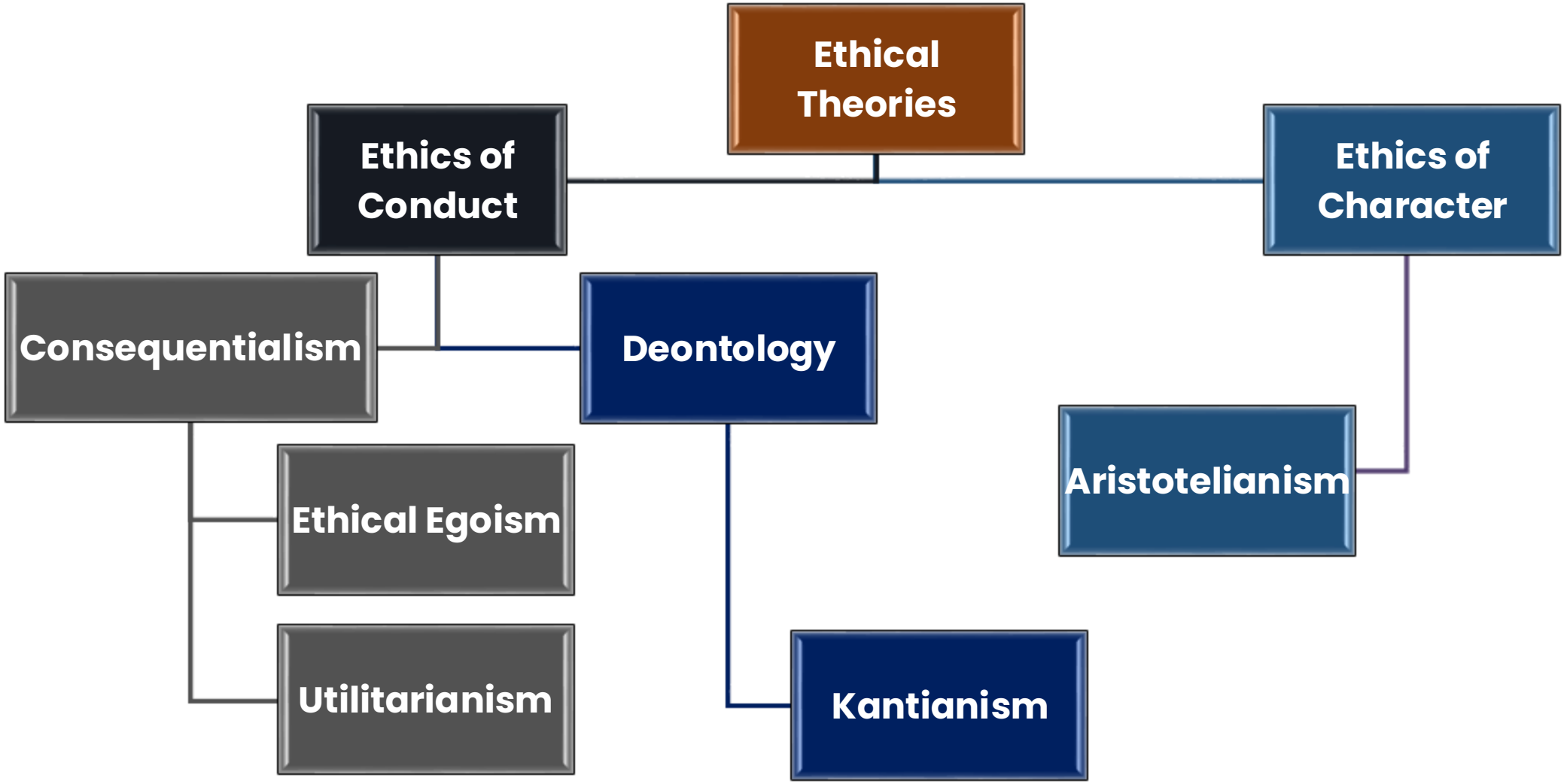
- Logical, Clear and coherent theory
- Internally consistent
- Theory and defence based in facts
- Basic moral values systematic and comprehensively organized (for prioritization)
- Compatibility with moral convictions

Theories: Introduction

- Theories and judgements must be constantly adjusted against each other (for the public good) so as to reach a consensus and avoid controversy
- All theories converge, usually in the welfare of the humanity, and reinforce the ethics
- Theories are usually complementary and not contradictory to each other

Naagarazan, R.S., "A Textbook on Professional Ethics and Human Values", New Age International Ltd., 2006

Classification of Ethical Theories



http://www.trinity.edu/cbrown/intro/ethical_theories.html

Ethical Theories: A Comparison

Consequentialism

- *e.g.* Utilitarianism
- Action is right if:
  - it promotes best consequences
- "Good" is "right"
  - "Ethical Egoism" for the agent
  - "Utilitarianism" for everyone affected

Deontology

- *e.g.* Kantianism
- Action is right if:
  - it follows a moral rule/ principle
- "Good" and "right" are independent of each other
  - Good is right only if it is according to supreme principle of morality

Virtue Theory

- *e.g.* Aristotelianism
- Action is right if:
  - it is what a "virtuous agent" would do in those circumstances
- Virtue is important

http://www.trinity.edu/cbrown/intro/ethical_theories.html

Ethical Theories: A Comparison

Consequentialism

- How do I get what I want/what is good?
- Will & Reason, Desires
- What is to be done?
- Consequences are important

Deontology

- What is rational?
- What is to be done?
- Will & Reason
- Acts are important

Virtue Theory

- What habits must be developed?
- How should one be?
- Will & Reason, Desires, Character, Traits
- Character/ Agents are important

http://www.trinity.edu/cbrown/intro/ethical_theories.html

Ethical Theories

Theories of Right Action

Deontological Ethics

Kant, Rawls (and Ross)

Duty Ethics Theory: Kant

- Actions are consequences of performance of one's duties – Immanuel Kant (18th c.)
  - Based on respect for autonomy
  - One has duties to oneself as one is autonomous and rational
  - Moral duties are "*categorical imperatives*" – commands imposed on oneself & others

Duty Ethics: Kantian Principles & Prescription

- Maxim: "I will <act> in certain <circumstance> in order to realise or produce <end>."
- An act in certain circumstance is morally permissible thus:
  - Step 1: Formulate the maxim in each act and circumstance
  - Step 2: Consider if this can be true in case of not just you but anyone
  - Step 3: Is it okay if everyone can act thus in that circumstance?
  - Step 4: So, can you rationally will (determine) to act as per the maxim?
- Duty is "perfect" if all steps pass and "imperfect" if Step 3 fails
  - Imperfect duties lead to exceptions for the act and circumstance

https://plato.stanford.edu/entries/kant-moral/

Duty Ethics Theory: Rawls

- Action is based on impartiality and voluntary consensus – John Bordley Rawls (20th c.)
  - Considers autonomy for all those involved and not just an individual decision-maker
  - Each person is entitled to the most extensive amount of liberty compatible with an equal amount for others – "maximal + equal liberty" ("*Liberty Principle*")
  - Differences in social power and economic benefits are justified only when they are likely to benefit everyone, including members of the most disadvantaged groups - "inequalities for common advantage" ("*Inequality Principle*")

Duty Ethics Theory: Ross

- Action must be based on specific guidelines defining prima facie duties – W.D. Ross (19th c.)
  - Duties may have justified exceptions or be "prima facie duties" – intuitively obvious
  - Moral Convictions
    - 1. Fidelity: Keep promises
    - 2. Reparation: Compensate others when we harm them
    - 3. Gratitude: Thank those who help us
    - 4. Justice: Recognize merit
    - 5. Beneficence: Recognize inequality and improve the condition of others
    - 6. Self-improvement: Improve virtue and intelligence
    - 7. Non-malfeasance: Do not injure others
- Criticism: Intuition may not provide sufficient guideline for moral duty

Duty Ethics: Summary

- Kantian Ethics
  - Is an action based on a “universalizable maxim” (moral rule) and is in accordance to principles that govern duty?
- Ross Ethics
  - Is an action based on a prima facie duty OR is an action based on intuition?
- Rawls Ethics
  - Is an action based on Maximum Equality or on Consensus regarding Inequality?

Virtue Ethics

Aristotelianism

Virtue Theory: Aristotle

- Character is more important than rights and duties
  - One's virtues make one's character
- Balance extremes of conduct, emotion, desire, attitudes – Aristotle
  - "Golden mean between extremes of excess and deficiency"
- Do what can achieve common good and social good (justice, health, safety, usefulness) – MacIntyre
  - See "Professional Virtues": Self-governance Virtues, Public-spirited Virtues, Team-work Virtues, Proficiency Virtues, Cardinal Virtues

Virtue Theory: Example

Virtue	Excess	Golden Mean	Deficient
Truthfulness (Communication)	Dishonouring confidentiality	Necessary and sufficient, to proper person(s) only	Secretiveness
Courage (Facing dangers, risk)	Rough, bold	Firm, humble	Cowardice
Generosity (Giving)	Wastage of resources	Give only as much needed, to the deserving	Miserly
Friendliness (Relationships)	No anger, unreserved	Within decent (and professional) limits	Bad-tempered
Green Environment	Exploitation	Protection	Negligence
Work and earn	Tiresome work	Balance of work and pleasure	No work, more pay

Naagarazan, R.S., "A Textbook on Professional Ethics and Human Values", New Age International Ltd., 2006

Consequentialism

Act & Rule Utilitarianism

Utilitarian Theory

- Jeremy Bentham & John Stuart Mill (19th c.) – to determine which laws were morally the best: Standard of right conduct is maximisation of good consequences ("utilities" or "balance of good over evil")
- Right action: Brings greatest satisfaction of preferences for those involved
  - Identify various courses of action
  - Who will get affected (benefited or harmed) by each action?
  - Choose the action based on greatest good for greatest number of people with least harm

Utilitarian Theory

- Act Utilitarian Theory (J.S. Mill):
  - Focuses on actions rather than on rules
  - Action is right if it generates overall good for most people involved

- Rule Utilitarian Theory (Richard Brandt, 1959):
  - Focuses on rules
  - Action is right if it is required by a set of rules which maximize public good

Utilitarianism: Approaches through Thinking

Descriptive/ Comparative

- Describes how people behave and moral standards they claim to follow
- Decisions are chosen by Pain or Pleasure
- "Practical"

Normative

- Defines how people should be and what people should do – creates or evaluates moral standards
- Morality enhances well-being and reduces suffering
- "Ideal"

Analytic/Meta

- What a society decides to be right is right and what they claim to be wrong is wrong – morality is relative
- Morality is a system to keep humans happy and alive
- "Reverse Engineering"

https://www.learnreligions.com/ethics-descriptive-normative-and-analytic-4037543

Utilitarian Understanding of Ethics

- What may be right in one society/ community/ context is wrong in another ("Ethical Relativism")
- Descriptive Ethics are merely observation without any conclusions
- Prescriptive/ Normative Ethics are based on observation but also conclude what's treated right or wrong in each different context, to define the norm (*i.e.*, to prescribe)
- Analytic Ethics *define* ethics based on observation

Ethics through God, Religion, and Spirituality

Divine Command Ethics

- Ethics commanded by the Word of Good or Book of Religion or a priest or other proponents of the Divine Command
  - Holy Bible
  - Qur'an
  - Ten Commandments
  - Bhagavad-Gita
  - Vedas-Upanishads
  - Avesta
  - Guru Granth Sahib
  - Torah/ Tanakh
  - …

Morals and Religion

- TED Talk by Frans de Waal
  - Moral Behaviour in Animals: https://www.ted.com/talks/frans_de_waal_moral_behavior_in_animals/transcript
  - "Evolved Morality": Based on empathy, consolation, pro-social tendencies, reciprocity, and a sense a fairness
- Pillars of Morality: Reciprocity (Justice & Fairness), Empathy/ Compassion (Body/ Emotional Channel & Cognitive/ Perspective Channel)
- Big Think Talk by Frans de Waal
  - Morality without Religion: https://www.youtube.com/watch?v=sEQuIDqY6Cc

Ethical Pluralism

- Pluralism: When more than one value system can be considered correct and fundamental but when the value systems contradict one another

- Examples:
  - Being a nun is noble. Being a mother is noble too.
  - Which religion to follow? Which philosophy to subscribe to? Which God to worship?

Justice/ Fairness Theory

- Equals must be treated equally and unequals unequally – Aristotle
- Is a moral (being applied) fair?
- Is there a fair treatment without discrimination or bias?
  - Bias/ Favouritism: Benefits some people without justifiable reason
  - Discrimination: Imposing burden on some people while some other similar ones do not face the same

Rights Theories

Kant, Locke, Melden, …

Rights Theory

- Right: "Entitlement to act" or "to direct other individual's act"
- Protective barrier for individuals from unjustified moral infringement
  - Every right comes with a duty of non-interference
  - "Rights that do not flow from duty well performed are not worth having."
  - "Truth resides in every human heart, and one has to search for it there, and to be guided by truth as one sees it. But no one has a right to coerce others to act according to his own view of truth." – M.K. Gandhi
- Human Rights are the Ultimate Appeal or Moral Bottom Line
  - Moral authority to make legitimate demands on others to respect our choices
    - Others can make similar claims on us
- Powerful foundation for Professional Engineers

Naagarazan, R.S., "A Textbook on Professional Ethics and Human Values", New Age International Ltd., 2006

Rights Theory: Immanuel Kant

- Kant's Respect for autonomy extends to Rights Theory also
  - Ability to choose one's actions freely
  - Right to retain the respect of those choices
  - Right to access the truth
  - Right of Privacy
  - Right to not be injured or violated
  - Right to what is agreed (in a contract)

Rights Theory: John Locke

- Action must respect everyone's moral rights – John Locke (17th c.)
  - Wrong if individual's rights are violated
  - Reflected in the declaring the Rights to Life, Liberty, and Pursuit of Happiness (by Thomas Jefferson, President of the USA, who wrote the Declaration of Independence in 1776)

Rights Theory: Abraham I. Melden

- Action must not harm other's life, health, liberty, or property – Abraham Irving Melden (20th c.)
  - Nature makes that mandatory!
  - Rights must be based on social welfare system

Naagarazan, R.S., "A Textbook on Professional Ethics and Human Values", New Age International Ltd., 2006

Rights Theory: Human Rights

- Liberty/Moral Rights
  - Rights are natural (or invented/created/granted by government)
  - Rights are universal
  - Rights are equal irrespective of caste, race, creed or sex
  - Rights are inalienable, i.e. one's rights cannot be transferred/sold to another

- Welfare Rights
  - Rights to benefit the needy for a decent human life (when deprived of those benefits)

Rights Theory: Economic Rights

- Consumer Rights[1]
  - Right to Information
  - Right to Safety
  - Right to Choice
  - Right to be Heard
  - Right to Redressal
  - Right to Consumer Education

- *"If you are a merchant or lawyer, you have then too, a duty to discharge towards your customers and clients from whom you derive a considerable pecuniary advantage."*

— M.K. Gandhi[2]

References: 1. Nagarazan, R.S., "A Textbook on Professional Ethics and Human Values", New Age International Ltd., 2006

2. "The Collected Works of Mahatma Gandhi", Vol. 1 (1894)

Ethical Framework

Ethical Framework: Overview

- An organisation must establish its Standard Operating Procedures (SOP), Codes of Ethics and Conduct, and other policies based on certain framework.
- Morals, Values, and Ethics must be its foundation.
- Organisations may develop their own frameworks while carefully avoiding ambiguity and conflict. (Ref. Theories of Right Action)

Ethical Framework – FAIR (Rowson, 2006)

- To work more effectively, earn trust, mutual support and respect, and how it can foster democratic ideals in the workplace and community.
  - Fairness
  - Respect for Autonomy
  - Integrity
  - Seeking most beneficial and least harmful consequences or Results

Rowson, R., "Working ethics: how to be fair in a culturally complex world", Jessica Kingsley Publishers, 2006

Ethical Framework – i.HOPE (D.Light)

- Social enterprise that provides off-grid, affordable solar-powered solutions to over two billion people in the developing world without access to reliable energy
  - Innovative
  - Honest
  - Optimistic
  - Passionate
  - Empathetic

http://www.dlight.com/about-us/

Ethics from Gita and Philosophies

Lessons to Learn from Bhagavadgita

- Do your duty without shunning away from it!
- Your duty does not define you! But your character defines how you perform your duty and vice versa
- Do not let the lack of control on outcome deter you! You have control on your actions!
- You have all the freedom in action, but not freedom from action!
- You must necessarily perform your action with full consciousness that the outcome cannot drive you towards it! Consequences may seem bad, but duty is unavoidable
- You are accountable; you have both causal responsibility and answerability for the consequence of your action
- Develop the necessary aptitude to perform your duties!

Oppenheimer and the Gita

- Oppenheimer called the Bhagavad Gita “the most beautiful philosophical song existing in any known tongue.” and was so inspired by it that he wrote to his brother it was “very easy and quite marvellous.”
- He formulated a code for living based on sacred literature of India and saw three principles of life, inspired by the literature: duty, fate, and faith
- He believed that he had a job to do; that he should do it only because it was his job and not because he was intent on obtaining any particular result; and that following these principles would bring a saving measure of peacefulness into his unhappy existence.
- Scientists should shun the ivory tower and act selflessly but effectively in the world, he believed, inspired by the Gita.
- Soon after Hiroshima and Nagasaki were obliterated, he became a leading critic of nuclear weapons and nuclear war. He said that the physicists who built the atomic bomb had “known sin”; that he himself had blood on his hands. Not only did he build the bomb, but to the end of his life—even after acknowledging that he had blood on his hands—he maintained that he had done the right thing.

Hijiya, J.A., “The Gita of J. Robert Oppenheimer”, Proceedings of the American Philosophical Society, 144(2):123–167 (2000)

From Eastern and Greek Philosophies



- *oubaitori*: Do not compare yourself! (Japanese)
- *saikata ga nai*: Accept and let go! (Japanese)
- *gaman*: Endure with patience and dignity! (Japanese)
- *wabi sabi*: Imperfections are okay! (Japanese)
- *sacchidänanda*: bliss of truth consciousness (Indian)
- *Eudaimonia*: virtuous life of good spirit (Greek)

Ethical Inquiries

Types of (Ethical) Inquiry

- Help ethical decision-making
- Complementary and inter-related
- All three types of inquiries may be required in every case
- Normative: To identify the correct way of doing something (*"How is something supposed to be done?"*)
- Conceptual: To clarify definitions, concepts, principles, and issues in engineering ethics (*"What is what?"*)
- Factual: To describe current practices and facts of business pertaining to Value issues (*"What exactly has been done or is to be done?"*)

Ethical Inquiries

Normative

- What must one do under certain circumstances?
  - Expected ideal, and does not depend on one's belief of right or wrong
- Morally desirable nature for guidance to individuals or groups
  - How can engineers protect public safety?
  - What are the laws and procedures affecting engineering practice on moral grounds?
  - How can moral rights be implemented to fulfill professional obligations of engineers?

Conceptual

- What does the concept/ principle mean/ describe?
- What ethics must engineers possess to protect safety, health, and welfare of the public?
- Compliance for guidance to individuals or groups
  - What is safety?
  - What are marginal issues of safety and what are the necessary precautions to avoid risk?
  - How can moral corruption can affect professionalism?

Factual

- What actually happened? How can one deduce the facts scientifically?
- Information for guidance to individuals or groups
  - What are the business realities of engineering practice?
  - What guidance may professional societies provide in the specific circumstance?
  - What are the necessary risk assessment procedures to be followed?
  - What are the ways to profile engineers psychologically?

Decision-Making Tools

Decision Making

- Prescriptive
  - How must optimal decisions be made?
- Descriptive
  - How do people make decisions?

https://en.wikipedia.org/wiki/Decision_analysis

Decision Making Methods

Individual

- Simple Prioritisation
- Decision Balance Sheet
- Satisficing
- Maximising
- Following Orders
- Anti-authoritarianism
- Automated/Software-based

Group

- Consensus
- Voting
- Delphi Method
- Collaborative/ Participative
- Decision Engineering
- Decision Conferencing

https://en.wikipedia.org/wiki/Decision-making#Decision-making_techniques

Decision Making Steps

GOFER

- Goals: Survey values and objectives.
- Options: Consider a wide range of alternative actions.
- Facts: Search for information.
- Effects: Weigh the positive and negative consequences of the options.
- Review: Plan how to implement the options.

DECIDE

- Define the problem
- Establish/ Enumerate all the criteria (constraints)
- Consider/Collect all the alternatives
- Identify the best alternative
- Develop and implement a plan of action
- Evaluate and monitor the solution and examine feedback when necessary

https://en.wikipedia.org/wiki/Decision-making#Steps

Decision Making Steps: Group

- Group-DM Stages (B Aubrey Fisher)
  - Orientation: Know
  - Conflict: Resolve
  - Emergence: Get clarity
  - Reinforcement: Reasoning and decision-making

https://en.wikipedia.org/wiki/Decision-making#Steps

Victor Vroom's Normative DM Model

- Decide
  - Information is collected from individuals – leader makes the decision
- Consult
  - Information is obtained in private or in a group meeting and decision is made accordingly
- Facilitate
  - Participative, Cooperative, Collaborative approach with leader being non-directive
- Delegate
  - Leader passes on DM to the group

https://en.wikipedia.org/wiki/Group_decision-making

Decision Matrix

- *M* options, *N* alternatives → M × N matrix

	Alternative 1	Alternative 2	...	Alternative *M*
Criterion 1	x_{11}	x_{12}	...	x_{1M}
Criterion 2	x_{21}	x_{22}	...	x_{2M}
...	...	...	X_{ij} = Good	...
Criterion *N*	x_{N1}	x_{N2}	...	x_{NM}
Sum				
Rank				
Status		No		No

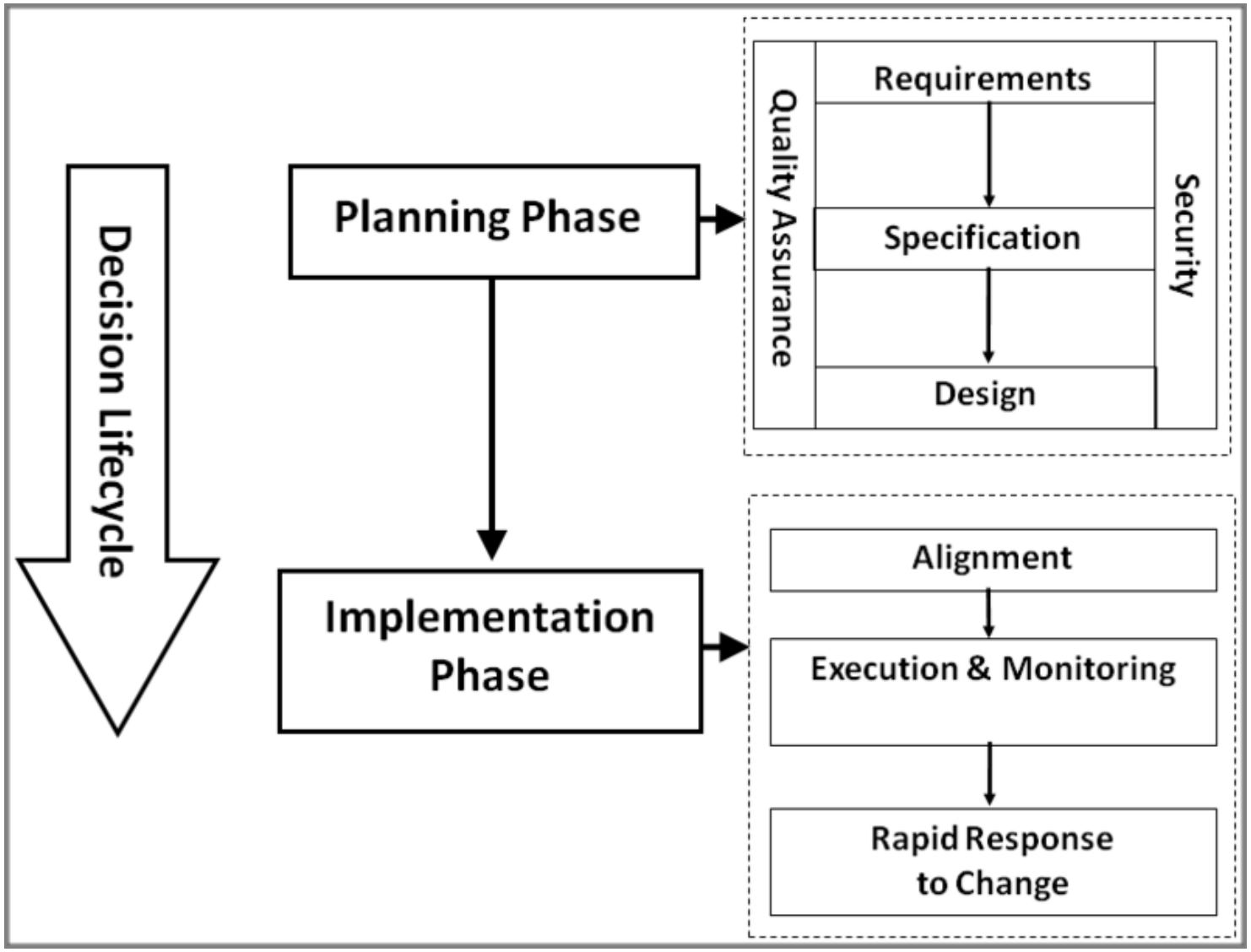
https://en.wikipedia.org/wiki/Decision_matrix

Decision Engineering

- Cognitive bias
- Decision making
- Situational awareness
- Critical and creative thinking
- Collaboration and organizational design, with engineering technologies
- Operations Research
- Engineering Principles
  - Requirements analysis
  - Specification
  - Scenario planning
  - Quality assurance
  - Security
- Design Principles

https://en.wikipedia.org/wiki/Decision_engineering

Decision Engineering/Intelligence



By Prazan - Own work, Public Domain, https://commons.wikimedia.org/w/index.php?curid=5218695

Decision Conferencing

- Engages key players from diverse perspectives in solving issues of concern (Dr. Cameron Peterson *et al.*, 1970s; L.D. Phillips, 1981)[1]
  - Developing a shared understanding of the issue
  - Creating a sense of common purpose
  - Generating a commitment to the way forward
- Four phases[2]
  - Broad Exploration
  - Model Construction
  - Extensive Exploration → Iterate
  - Action Plan/Recommendations
- Organisation's Culture[2]
  - Participative DM
  - Openness to change
  - Climate for problem-solving
  - Distributed authority and accountability

1. https://en.wikipedia.org/wiki/Decision_conferencing

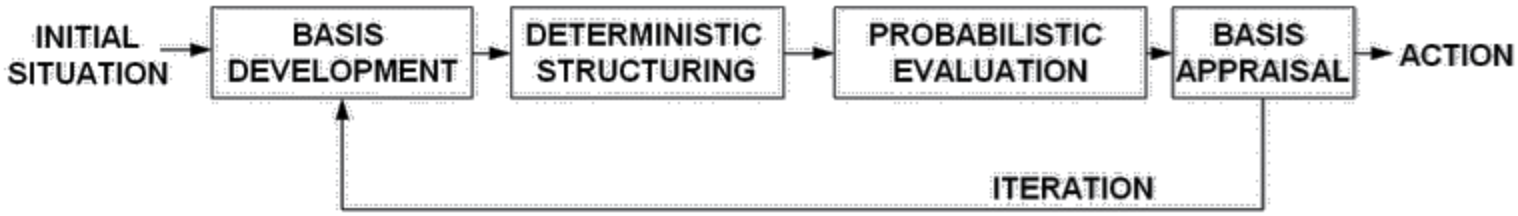
2. http://www.lawrencephillips.net/decision-conferences.html

Decision Analysis (DA)

- Comprises of philosophy, theory, methodology, and professional practice
- Procedures, methods, and tools for identifying, clearly representing, and formally assessing important aspects of a decision
- Applies Maximum expected utility action

https://en.wikipedia.org/wiki/Decision_analysis

DA Cycle



By Ronald Howard - http://decision.stanford.edu/library/ronald-a.-howard/the-evolution-of-decision-analysis-1, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=7813103

Decisions through Engineering

- Decision Trees
- Operations Research (ME)
- Decision-Making Software (CSE)
- Decision Support System (IT)
- Knowledge-Based/ Adaptive Systems
- Artificial Intelligence

Ethics and Law

Ethics and Law

- Ethics focus on dos. Law emphasizes on don'ts.
- Ethics and Law go hand in hand ideally, but they are independent of each other – certain acts can be legal but not ethical or ethical but not legal too!

- Ethical action is that which best serves the ideas of honesty, integrity, morality and good management practices.
- Law can be defined as a consistent set of universal rules that are widely published, generally accepted, and usually enforced.

- *Equals should be treated equally and unequals unequally in proportion to the relevant inequalities.*

— Aristotle

- *There is only one ethics, one set of rules of morality, one code that of individual behaviour in which the same rules apply to everyone alike.*

— Peter F. Drucker

Paliwal, M., "Business Ethics", New Age International, 2006; http://amr.aom.org/content/20/2/379

Rights of Criminals and Convicts

Geneva Conventions (1949)

- International humanitarian laws defining basic rights of wartime prisoners, civilians and non-combatants, and military personnel – Ratified (accepted) by 196 countries
- First Geneva Convention: for the Amelioration of the Condition of the Wounded and Sick in Armed Forces in the Field (first adopted in 1864, final revision in 1949)
- Second Geneva Convention: for the Amelioration of the Condition of Wounded, Sick and Shipwrecked Members of Armed Forces at Sea (first adopted in 1949)
- Third Geneva Convention: relative to the Treatment of Prisoners of War" (first adopted in 1929, final revision in 1949)
- Fourth Geneva Convention: relative to the Protection of Civilian Persons in Time of War (first adopted in 1949)
- 1951 Refugee Convention: Refugees must abide by the law of the land where they seek asylum (refuge) and the contracting state (asylum nation) shall not discriminate against refugees and provides them certain rights equal to their own nationals (religion, education, IP, public relief, rations, labour legislation, social security)
- Hateful treatment of criminals in jails, encounters leading to death, *etc.* are illegal! (Refer: Ajmal Kasab, Afzal Guru, terrorists, naxals and supporters)

https://en.wikipedia.org/wiki/Geneva_Conventions https://en.wikipedia.org/wiki/Convention_Relating_to_the_Status_of_Refugees

Provisions in Constitution of India and CrPC

- Right to know the ground of arrest
- Right to be produced in front of magistrate without delay
- Access to bail
- Right to fair trial
- Right to free legal aid
- Right to be examined by doctor
- Wrong arrests may receive compensation (up to ₹1000)
- Bharatiya Nagarik Suraksha Sanhita, 2023 replaced Criminal Procedure Code (CrPC)
  - Specifies timelines for how long a case can be pending, right from framing of charges to pronouncing the verdict — to protect rights of the convicts too!

https://timesofindia.indiatimes.com/readersblog/your-life-your-decisions/what-are-the-rights-of-the-accused-person-in-india-37467/
https://www.lawyersclubindia.com/articles/rights-of-the-accused-person-under-crpc-15354.asp